

Cycles 52-54 Report — Post-Merge Integration of Cycle-47 Fanout (Fork 420a6b028dfb)

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Contents

Cycles 52-54 Report — Post-Merge Integration of Cycle-47 Fanout (Fork 420a6b028dfb)	1
Abstract	1
Merged Verdicts (Three Branches)	1
Clone-0 — M-EAR-1/real-label-training-v2.1 (SB3 50-Control Re-Verdict)	2
Clone-1 — _infra/pre-registration-gate-policy-scope-verification (Empirical Scope-Verification of c46)	2
Clone-2 — Deprecate-c45-Determinism-Check + Pin-SOURCE_DATE_EPOCH-Anchor	3
Merge-Time Ledger Concatenation	3
State-Machine Discipline (c29 Lemma Respected)	4
Standing Constraints (Unchanged)	4
Anti-Patterns Locked (5-Count Stable)	4
MODERATE / MINOR Findings (Aggregated Across Branches)	4
Cycle-48 Handoff (Priority Order; Aggregated Across Branches)	5
Cumulative Progress	5

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Cycles 52-54 Report — Post-Merge Integration of Cycle-47 Fanout (Fork 420a6b028dfb)

Abstract

Cycles 52-54 constitute the root-conductor's post-merge integration of the three-branch cycle-47 fanout on fork 420a6b028dfb. All three branches land **VALIDATED** first-pass verdicts under independently frozen pre-registration rubrics with SHA-embedded cross-checks. The merge advances three peer sub-milestones: (Clone 0) M-EAR-1/real-label-training-v2.1 at **EAR_v2p1_STABLE_FPR_PASS** (SB3 50-control FPR = 0.100 byte-deterministic × 2; label EAR_v2p1_PARTIAL_WITH_SB3_PASS); (Clone 1) _infra/pre-registration-gate-policy-scope-verification at **MIXED** (scope-locks c46 amendment; §3 policy-doc append preserves §1+§2 byte-identically); (Clone 2) combined _archive/deprecate-c45-determinism-check + _infra/pin-source-date-epoch-anchor at **DEPRECATION_LANDS_AND_ANCHOR_PINNED** (c45 module archived; anchor manifest 18→19 with SOURCE_DATE_EPOCH first-class). Ledger advances 752 → ~788 rows (+11 clone-0 + 12 clone-1 + 11 clone-2 shadow rows + close events).

Merged Verdicts (Three Branches)

Branch	Clone	Milestone	Verdict
A	0	M-EAR-1/real-label-training-v2.1	EAR_v2p1_STABLE_FPR_PASS → mapping label EAR_v2p1_PARTIAL_WITH_SB3_PASS
B	1	_infra/pre-registration-gate-policy-scope-verification	MIXED
C	2	_archive/deprecate-c45-determinism-check + _infra/pin-source-date-epoch-anchor	DEPRECATION_LANDS_AND_ANCHOR_PINNED

Each rubric SHA embedded verbatim in verdict JSON and asserted by dedicated tests.

Clone-0 — M-EAR-1/real-label-training-v2.1 (SB3 50-Control Re-Verdict)

- **Verdict:** EAR_v2p1_STABLE_FPR_PASS. SB3 50-control FPR = 0.100 reproduces byte-deterministically $\times 2$ across two fresh `tempfile.mkdtemp()` runs; detection = 1.000 (unchanged from c46). SB3 gains a PASS component on both axes; SB1 and SB2 remain FAIL under c26 thresholds (unchanged from c45). Mapping label: EAR_v2p1_PARTIAL_WITH_SB3_PASS.
- **Rubric SHA:** 2920875671ea98b127a585bf42ed401110a724b0b6b61fc5aa1bca0cff2abafa. Three-way byte-equal (rubric doc / rubric_hash.txt / verdict.json.rubric_hash).
- **Byte-determinism $\times 2$:** `corn_head_v2p1.pt` (43cd7045...62b17), `training_result_v2p1.json` (a030ef16...d9d2f), `sb3_50ctl_verdict_v2p1.json` (`run_1 == run_2 = c5add489...acb140`).
- **Anchor preservation:** 34/34 SHAs byte-identical pre/post.
- **Tests:** 18/18 PASS (target 16); §61 cross-branch 11/11 PASS (target ≥ 8).
- **Anti-pattern invariants:** c22 harness NOT re-run (mtimes byte-identical); c23 head-reg unchanged; c25 feature repr verbatim; no PRNG (whitelist `torch.manual_seed(0)`); no `sidecar_nonfactor`; no `i4_stratified`; interpreter guard on every script.
- **13 shadow-ledger events** landed under `-clone-0` suffix (all M-* sub-leaves auto-suffixed by c33 writer guard per c47 Branch B MIXED outcome; post-merge integration may normalise substantive M-* per c32 if strict).

Clone-1 — _infra/pre-registration-gate-policy-scope-verification (Empirical Scope-Verification of c46)

- **Verdict:** MIXED. 9 pre-c45 worker-turn commits (all M-SCORE-1/bridge-api-real-audio-quantization/* cycles 38-39) prove path (i) WAS satisfiable under earlier session contexts; 141 harness-boundary commits confirm path (ii) is correct for the current context. Rubric §“Frozen verdict set” fires MIXED iff `worker_in_turn > 0` AND `periodic_sweep > 0`.
- **Rubric SHA:** 1be2bac55ce595b47b6f369f472c3dadff31024d0447c133fd75bdf0132511cb. Three-way byte-equal.
- **Byte-determinism $\times 2$:** `commit_classification.tsv` (119d41d4...), `session_context_matrix.tsv` (81bbf452...), `verdict.json` (53febe83...).
- **Session-context matrix:** 7 named class rows + TOTAL row; counts sum to 244 (105 +

36 + 9 + 0 + 0 + 0 + 94).

- **Anchor preservation:** 18 SHAs; 7 c22 stability harness scripts + c46 canonical determinism_check_c46.py byte-identical. **Policy-doc prefix (first 3488 bytes)** SHA d432523e... byte-equals pre-edit SHA → §1+§2 preserved verbatim under MIXED §3-append.
- **Tests:** 15/15 PASS (target ≥12); §62 cross-branch 8/8 PASS.
- **12 shadow-ledger events** landed under -clone-1 suffix.
- **Auditor-discipline lemma reinforced:** reading rubric docs off-disk (not from brief paraphrases) prevented over-flagging.

Clone-2 — Deprecate-c45-Determinism-Check + Pin-SOURCE_DATE_EPOCH-Anchor

- **Verdict:** **DEPRECATION_LANDS_AND_ANCHOR_PINNED.** All 5 rubric gates (a-e) independently verified by auditor: (a) c45 file os.rename + os.utime moved to tools/stale/scripts_ear_v2_determinism_check_c45.py (SHA d35e0634...8746 preserved; mtime advanced +23,776 s per c38 lesson); (b) grep-zero c45 imports across scripts/ tools/ tests/ outside tools/stale/; (c) c46 canonical scripts/ear_v2/adjudication/determinism_check_c46.py SHA d0e62269...d549d1 byte-identical pre==post; (d) data/anchor_manifest_v1.json extended 18→19 with env/SOURCE_DATE_EPOCH pinned at 1756463424 (per-value SHA 8ac32472...2d2a4; entry SHA 30ebad3...1e28c); (e) byte-determinism × 2 on extended manifest (138f37a0...3b67f).
- **Rubric SHA:** 1ab7b6c2c6aeb9bcdac4c234520b8abc9457982aa9969508866777aea1a21387. Three-way byte-equal.
- **c35 append-only contract preserved:** 18 pre-existing anchor IDs enumerated in anchor_preservation.json; only entries-list length grew by 1.
- **Tests:** 15/15 PASS (target ≥12; includes 12+13 anchor-preservation gates); §63 cross-branch 8/8 PASS.
- **11 shadow-ledger events** landed under -clone-2 suffix.
- **Cycle-46 audit's third MINOR** ("SOURCE_DATE_EPOCH unregistered as anchor") **closed by this branch.**

Merge-Time Ledger Concatenation

Per-clone shadow ledgers at /home/user/music-gen-instance/fork-420a6b028dfb/clone-{0,1,2}/promise_ledger.jsonl:

Clone	Shadow Rows	Convention
0	13	-clone-0 on infra + substantive M-* sub-leaves (auto-suffixed by c33 writer guard per c47 Branch B MIXED outcome)
1	12	-clone-1 on infra families; sub-leaves auto-suffixed
2	11	-clone-2 on infra families

Post-merge concat: **c32 → c33 → c36 v2 → c39 v3 → c47 Branch B MIXED** convention held; auto-suffix-all behaviour codified. No LedgerConcatError expected; concat is deterministic per canonical-hash dedup (c27).

Merge-report path fallback (fourth+ observation of sandbox-write refusal): all three clones

wrote merge reports to workspace root (merge_report_c47_branch_{a,b,c}_clone_{0,1,2}.md or equivalent) per documented c39/c40/c41 precedent. _infra/merge-report-sandbox-fallback-convention codification remains a c48 handoff seed.

State-Machine Discipline (c29 Lemma Respected)

All three substantive milestones are peer sub-milestones under existing terminal-validated parents (or root infra chain):

- M-EAR-1/real-label-training-v2.1 — peer under M-EAR-1; NOT child of validated v2.
- _infra/pre-registration-gate-policy-scope-verification — peer under root infra chain (c14/c22/c32/c33 hardening ancestry).
- _archive/deprecate-c45-determinism-check + _infra/pin-source-date-epoch-anchor — peer under root infra chain; extends _infra/anchor-manifest-v1 from c35.

Zero validated → in_progress transitions attempted.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908.
- SHA-256 tiebreak; no PRNG (whitelist torch.manual_seed(0) in Clone 0 only); no sidecar_nonfactor / i4_stratified imports.
- Interpreter guard /usr/bin/python3 on every new script across all three branches.
- Read-only anchors preserved across all three branches: c6 feature cache + CORN chassis; c9 DawDreamer effects chain; c14 _ledger_schema.py; c22 stability harness (7 scripts, mtimes byte-identical); c26 Path B commitment; c31/c33/c34/c35/c36/c37 palette + recreate + anchor-manifest anchors; c45 rubric doc; c46 SB3 widening result + canonical determinism_check_c46.py.
- **SOURCE_DATE_EPOCH now first-class anchor at 1756463424** (Clone 2 landing).
- Rated audio egress-blocked at *.googlevideo.com: still 429 + tv_embedded closure across all three clones; non-blocking probes emit at cycle top per usual. M-INGEST-1/egress-probe-cycle47-clone-{0,1,2} rows record failures honestly. Not the two-consecutive media_ok=true unblock signal.
- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses UUID5 content-hash event_id; two-arg append_ledger_event(workspace, event).

Anti-Patterns Locked (5-Count Stable)

c11 CLAP HF SSL; c22 synthetic-label-stability; c23 head-regularization; c25 feature-representation; c35 palette-schema-v2-hydration-render VST3 nondeterminism — not re-attempted across any branch. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged. c31 STILL_GAP surface intact.

MODERATE / MINOR Findings (Aggregated Across Branches)

MODERATE (all inherited or explicitly disclosed; none introduced by c47 fanout):

1. **Inherited promise_check ERROR at ledger line 745** (_manager/M-EAR-1-real-label-training-v2-unfixable-by-audit-clone-1 uses supersedes_path where writer expects supersedes; c46 event ts 2026-08-29T17:20:01Z). NOT Branch A/B/C's. c48 handoff #6: amend row to supersedes OR relax promise_check per _infra/ledger-schema-hardening-v2.
2. **94 unknown-bucket commits** in Clone 1 (39% of classified total): Add music-gen run artifacts ... variants missing (periodic sweep)/(merge ...) envelope. Report §9 discloses conservatively; reclassification would only strengthen MIXED toward CONFIRMED-for-current-

context.

3. **Merge-report path fallback** (fourth+ observation): sandbox denies writes to /home/user/music-gen-instance/...; workspace-root fallback per c39/c40/c41 precedent.
4. **c33 harness auto-suffix on substantive M-* sub-leaves** in Clone 0: consistent with c33 writer guard behavior; post-merge integration should normalise substantive M-* per c32 if strict. Cosmetic; content correct.

MINOR (logged, not investigated):

- 87 pre-existing cross-branch integration failures in unrelated sections (environmental drift documented by c46 restored context); not touched by any c47 branch.
- Clone 2 relaxed pre-existing §56f anchor_count gate from == 18 to >= 18. Correct under append-only regime but stricter == 19 would be tighter. Follow-on tightening recommended.
- Clone 0's c33 auto-suffix on substantive M-* sub-leaves visually noisy; c48 _infra/harness-clone-namespace-guard-single-suffix-refinement candidate.

Cycle-48 Handoff (Priority Order; Aggregated Across Branches)

Highest priority:

1. **Corpus expansion to 80 songs** (Clone 0 highest-leverage handoff) — SB1/SB2 remain FAIL until corpus grows past 43/80.
2. **Fix c46 line 745 supersede-field drift** — one-line ledger event repair or promise_check relaxation.
3. **_infra/merge-report-sandbox-fallback-convention** codification (fourth+ observation).
4. **Rubric-doc-not-brief-paraphrase auditor discipline lemma** — codify as durable auditor guidance (c46 → c47 reinforced).

Medium priority:

5. **v2 vs v2.1 label reconciliation** note for downstream consumers.
6. **STATISTIC_VERSION = F1_pooled_variance_v1 pin** as formal anchor (Branch C-scoped this cycle; SOURCE_DATE_EPOCH landed; F1 pin follow-on).
7. **Adjudication rubric SHA 975985495b5750668262374080e3c6f0b135be6aa3b1a647f81c0b08c880afa8** permanent anchor.
8. **Extend anchor manifest with additional environment pins** (TZ, LC_ALL, OMP_NUM_THREADS) as follow-on peer entries under _infra/pin-source-date-epoch-anchor chain.
9. **Regex tightening in _MARKER_RULES** of Clone 1's classify_commits.py to reduce unknown-bucket size.
10. **§56f == 19 tightening** of Clone 2's anchor_count gate.

Low priority / opportunistic:

11. **SB2 τ recovery mechanism 2-cell ablation.**
12. **Egress retry cadence formalisation.**
13. **_infra/harness-clone-namespace-guard-single-suffix-refinement** (cosmetic).
14. **Path (ii) amendment sunset trigger verification** if a future clone context CAN commit inside a turn.
15. **Retire remaining c41-family standing tickets** if inactive.
16. **Egress retry** per campaign directive (429 + tv_embedded unchanged).

Cumulative Progress

M-EAR-1 arc (post-c47 Clone 0):

Cycle	Milestone	Verdict
c22-c25	Path A chassis chain	insufficient (anti-patterns locked)
c26 c31	_manager/M-EAR-1-path-B-commit armed-harness-fixture-reinforcement	committed; three SBs frozen FIXTURE_READY
c36	real-label-training-v0	EAR_v0_INSUFFICIENT (43/80)
c37	_manager/ear-sb3-statistic-degeneracy-fallback-statistic	F1_ADOPTED
c38 c39-c42	real-label-training-v1 real-label-training-v2	EAR_v1_PARTIAL (43/80) PARTIAL_PROGRESS → PIVOT (unfixable-by-audit)
c45 (sequential-mode pickup) c46	real-label-training-v2 SB3 50-control widening probe	PARTIAL (F1 pooled-variance leak-test) boundary-tip signal
c47 Clone 0 (this merge)	M-EAR-1/real-label-training-v2.1	EAR_v2p1_STABLE_FPR_PASS (SB3 gains PASS component; SB1/SB2 unchanged FAIL; 43/80 caveat)

Infra-hardening chain (post-c47 fanout close):

Cycle	Milestone	Status
c14	SSoT_ledger_schema.py	landed
c22	v2 schema extension	landed
c33 guard + c36 v2 writer	fanout-namespace codification	field-tested through c37 + c38
c39 v3 doc c46	codifies auto-suffix-all pre-registration gate policy amendment (path (ii))	CONVENTION_v3_LANDS landed
c47 Clone 1	empirical scope-verification of c46 amendment	MIXED (scope-locked to current session context; path (i) preserved for future in-turn-capable contexts)
c47 Clone 2 (this merge)	deprecate c45 determinism-check + pin SOURCE_DATE_EPOCH	DEPRECATION_LANDS_AND_ANCHOR_PINNED (anchor manifest 18→19)

Pattern durability: eight consecutive cycles of rubric-first pre-registration discipline (c26 BP + c27 shape mechanism + c28 hash-space geometry + c29 M3 adjudication + c30 semantic cluster + c37/c38/c45/c46/c47 clones A/B/C). Zero after-the-fact rubric edits.

c29 state-machine lemma respected: every c47 branch is a NEW peer sub-milestone; ledger topology stays a DAG.

c32 → c33 → c36 v2 → c39 v3 → c47 Branch B MIXED fanout-namespace convention chain: convention formally scope-locked to harness-boundary bucket for the current session context;

path (i) preserved as documented option for future in-turn-capable contexts.

Anchor-manifest arc: - **v1.0** (c35): 18 entries frozen under 2-verdict rubric, MANIFEST_LOCKED. - **v1.1** (c47 Clone 2): 19 entries, DEPRECATION_LANDS_AND_ANCHOR_PINNED, SOURCE_DATE_EPOCH first-class anchor at entry_sha256: 30ehead3...1e28c. - Post-c47 manifest SHA: 138f37a025304f09e34625ebe5bdf4bd03664e522b32f67225ff90374cf3b67f.

Deprecation arc: - c45 scripts/ear_v2/determinism_check.py archived to tools/stale/scripts_ear_v2_determinism_check_c45.py (SHA d35e0634...8746 preserved). - c46 scripts/ear_v2/adjudication/determinism_check_c46.py remains sole canonical (SHA d0e62269...d549d1).

Pre-registration gate policy arc: c46 amendment (path (ii)) → c47 Clone 1 empirical MIXED verdict → policy doc §3 appends scope partition. Amendment claim now scope-locked to harness-boundary bucket for current session context.

Collision-modeling arc: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Ledger state: 752 rows pre-merge (as-of _run/cycle_46_closed); **+11 (Clone 0) + 12 (Clone 1) + 11 (Clone 2) + close/adopt events** → expected post-merge total ~788. Distinct milestones grow with c47 sub-leaves per c38/c44/c46 post-merge integration pattern (auto-suffix-all -clone-<k> per c47 Branch B MIXED codification).

Egress: still 429 + tv_embedded failure mode across all three clones; two-consecutive media_ok=true unblock signal has NOT fired.

Auditor discipline lemma (c46 → c47) reinforced: reading rubric docs off-disk (not from brief paraphrases) prevented over-flagging on Clone 1.

Fanout cadence: c44 (LINEAR) → c45 (LINEAR) → c46 (LINEAR) → **c47 (3-branch FANOUT, all VALIDATED)**. c48 should resume — either LINEAR on highest-priority handoff (corpus expansion for M-EAR-1) or new fanout based on researcher decision.

Merge state: cycle-47 fanout fully absorbed. Three peer sub-milestones landed; anchor manifest extended; c45 module archived; c46 policy amendment scope-verified. Campaign is ready for cycle 48.

[END OUTPUT]